MUSTAFA KURBAN

Robotics & Autonomous Systems Engineer

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PROFESSIONAL SUMMARY

Robotics and Autonomous Systems Engineer with 3+ years of hands-on experience designing, modelling, and deploying autonomous mobile and unmanned vehicle systems. Experienced in navigation and control algorithm development, sensor integration (LiDAR, IMU, Camera), and simulation environments including Gazebo, CARLA, and AirSim. Proficient in C++ and Python with a strong foundation in ROS/ROS2-based system architectures.

PROFESSIONAL EXPERIENCE

Software Engineer | Saha Robotic *Apr 2022 – Nov 2025*

- Developed and deployed indoor Autonomous Mobile Robots (AMRs) including cleaning and restaurant service robots using ROS and ROS2.
- Designed and implemented navigation and localization pipelines using Nav2, Cartographer, and SLAM Toolbox for real-world environments.
- Integrated and fused multi-sensor data (LiDAR, Camera, IMU) to achieve robust localization and obstacle avoidance.
- Led end-to-end development of a multi-robot fleet management system enabling centralized task assignment, real-time monitoring, and coordinated control of robot fleets.
- Built, validated, and stress-tested autonomous behaviours in simulation environments: Gazebo, CARLA, and AirSim, prior to hardware deployment.
- Performed on-robot system testing, debugging, and performance optimization in production settings.

ROS Robotics Instructor (University Bootcamp) | Saha Robotic — Baku, Azerbaijan *Part-time*

- Designed and delivered ROS and ROS2 bootcamp curriculum for university students.
- Taught core topics including coordinate transforms (TF), navigation stacks, SLAM, and sensor integration.
- Guided students through hands-on mobile robot simulation projects using Gazebo.

Software Engineer Intern | SK Robotic *Jun 2021 – Jul 2021*

- Migrated an autonomous forklift system into Gazebo simulation environment.
- Integrated robot models and sensor configurations into the ROS simulation stack.
- Supported testing of navigation and control algorithms during the internship period.

EDUCATION

Bachelor's Degree in Computer Engineering | Sakarya University *Aug 2019 – Aug 2023*

GPA: 2.77 / 4.00

CONFERENCES & TALKS

Speaker — *"Robotics and Sensor Physics: An Interdisciplinary Perspective"* *May 2026*

Sakarya University Physics Society, Sakarya, Türkiye

Invited as a guest speaker to deliver an interdisciplinary talk bridging robotics and sensor physics, covering the working principles of LiDAR, IMU, and camera systems, and their integration into autonomous mobile platforms.

TECHNICAL SKILLS

Programming Languages

- C++ — **Advanced**
- Python — **Advanced**

Robotics Frameworks & Tools

- ROS / ROS2 — **Advanced**
- Nav2 — **Advanced**
- Cartographer — **Advanced**
- SLAM Toolbox — **Advanced**

Simulation Environments

- Gazebo — **Advanced**
- CARLA — **Advanced**
- AirSim — **Intermediate**

Autonomous Systems

- Fleet Management & Multi-Robot Coordination — **Advanced**
- Sensor Fusion (LiDAR, IMU, Camera) — **Intermediate**

Other Tools

- MATLAB / Simulink — **Basic**

LANGUAGES

Turkish	Native
English	Professional Working Proficiency